

# Image quality metrics: PSNR vs. SSIM

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**Abstract**—In this paper, we analyse two well-known objective image quality metrics, the peak-signal-to-noise ratio (PSNR) as well as the structural similarity index measure (SSIM), and we derive a simple mathematical relationship between them which works for various kinds of image degradations such as Gaussian blur, additive Gaussian white noise, jpeg and jpeg2000 compression. A series of tests realized on images extracted from the Kodak database gives a better understanding of the similarity and difference between the SSIM and the PSNR.

**Keywords**—PSNR; SSIM; image quality metrics

## I. INTRODUCTION

Any processing applied to an image may cause an important loss of information or quality. Image quality evaluation methods can be subdivided into objective and subjective methods [1, 2]. Subjective methods are based on human judgment and operate without reference to explicit criteria [3]. Objective methods are based on comparisons using explicit numerical criteria [4, 5], and several references are possible such as the ground truth or prior knowledge expressed in terms of statistical parameters and tests [6-8]. In this paper we explicit the relationship between the SSIM and the PSNR for grey-level (8 bits) images. Given a reference image  $f$  and a test image  $g$ , both of size  $M \times N$ , the PSNR between  $f$  and  $g$  is defined by:

$$PSNR(f, g) = 10 \log_{10} (255^2 / MSE(f, g)) \quad (1)$$

where

$$MSE(f, g) = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N (f_{ij} - g_{ij})^2 \quad (2)$$

The PSNR value approaches infinity as the MSE approaches zero; this shows that a higher PSNR value provides a higher image quality. At the other end of the scale, a small value of the PSNR implies high numerical differences between images. The SSIM is a well-known quality metric used to measure the similarity between two images. It was developed by Wang *et al.* [9], and is considered to be correlated with the quality perception of the human visual system (HVS). Instead of using traditional error summation methods, the SSIM is designed by modeling any image distortion as a combination of three factors that are loss of correlation, luminance distortion and contrast distortion. The SSIM is defined as:

$$SSIM(f, g) = l(f, g)c(f, g)s(f, g) \quad (3)$$

where

$$\begin{cases} l(f, g) = \frac{2\mu_f\mu_g + C_1}{\mu_f^2 + \mu_g^2 + C_1} \\ c(f, g) = \frac{2\sigma_f\sigma_g + C_2}{\sigma_f^2 + \sigma_g^2 + C_2} \\ s(f, g) = \frac{\sigma_{fg} + C_3}{\sigma_f\sigma_g + C_3} \end{cases} \quad (4)$$

The first term in (4) is the luminance comparison function which measures the closeness of the two images' mean luminance ( $\mu_f$  and  $\mu_g$ ). This factor is maximal and equal to 1 only if  $\mu_f = \mu_g$ . The second term is the contrast comparison function which measures the closeness of the contrast of the two images. Here the contrast is measured by the standard deviation  $\sigma_f$  and  $\sigma_g$ . This term is maximal and equal to 1 only if  $\sigma_f = \sigma_g$ . The third term is the structure comparison function which measures the correlation coefficient between the two images  $f$  and  $g$ . Note that  $\sigma_{fg}$  is the covariance between  $f$  and  $g$ . The positive values of the SSIM index are in  $[0, 1]$ . A value of 0 means no correlation between images, and 1 means that  $f = g$ . The positive constants  $C_1$ ,  $C_2$  and  $C_3$  are used to avoid a null denominator.

There are no precise rules for selecting the SSIM or the PSNR when the evaluation of image quality is required. Consequently, informal arguments and belief guide the interpretation of numerical values obtained during the evaluation process [10-13]. In fact, some studies have revealed that as opposed to the SSIM, the MSE and so the PSNR perform badly in discriminating structural content in images since various types of degradations applied to the same image can yield the same value of the MSE [14]. Other studies have shown that the MSE, and consequently the PSNR, have the best performance in assessing the quality of noisy images [2]. The goal of this paper is to derive a simple analytical relationship between the SSIM and the PSNR that can be used to better understand their difference and similarity in the case of common degradations such as Gaussian blur, additive Gaussian noise, jpeg and jpeg2000 compression. We also compare in this paper the degree of sensitivity of the PSNR and the SSIM to those various degradations. In all of our study, we focus only on objective measurements and we do not address any subjective evaluation. The rest of the paper is organized as follows: in Section 2, we give a description of the derivation of an analytical relationship between the SSIM and the PSNR. In Section 3, we make a series of tests on natural images and we use some statistical models to compare the sensitivity of

the two quality measures to various degradations. We end the paper with concluding remarks.

To establish the relationship between the SSIM and the PSNR, we first derive the relationship between the SSIM and the MSE, and then we use that relationship to link the SSIM to the PSNR. The MSE in equation (2) can be rewritten as:

$$\sigma_f^2 = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N (f_{ij} - \mu_f)^2, \sigma_{fg} = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N (f_{ij} - \mu_f)(g_{ij} - \mu_g) \quad (6)$$

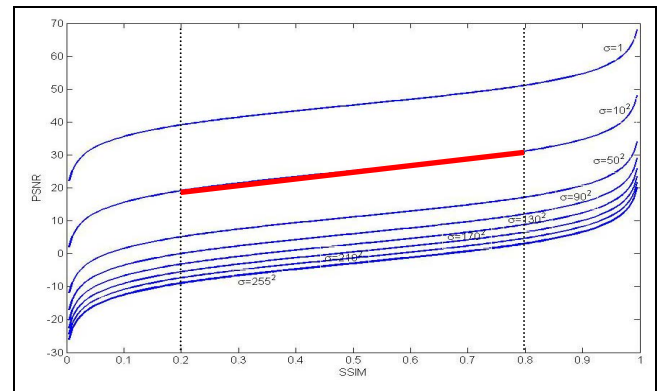
$$\frac{1}{SSIM} = \frac{255^2 \times \alpha(f, g) \times e^{-PSNR \times \ln(10)/10} + \beta(f, g)}{l(f, g)s(f, g)} \quad (7)$$

$$\begin{aligned} \alpha(f, g) &= \frac{1}{2\sigma_f \sigma_g + C_2} \quad , \quad \beta(f, g) = \frac{2\sigma_{fg} - (\mu_f - \mu_g)^2 + C_2}{2\sigma_f \sigma_g + C_2} \quad , \\ s(f, g) &= \frac{\sigma_{fg} + C_3}{\sigma_f \sigma_g + C_3} \end{aligned} \quad (8)$$

$$PSNR = 10 \log_{10} \left[ \frac{2\sigma_{fg}(l(f, g) - SSIM)}{255^2 SSIM} + \left( \frac{\mu_f - \mu_g}{255} \right)^2 \right] \quad (9)$$

Kodak database as well as blocks of size  $64 \times 64$  and  $16 \times 16$  within each of these images. We note that all the images were first converted into grey-level images before the computations take place. In overall, almost 6 000 000 computations of the luminance comparison function have been performed.

Figure 1. Images of the Kodak database



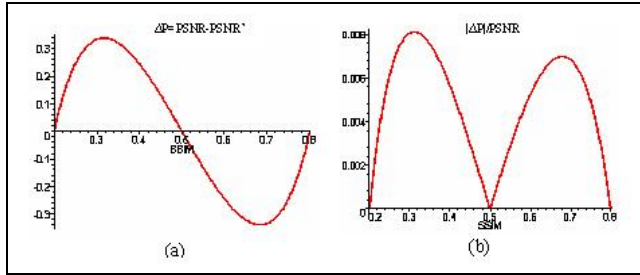


Figure 3. (a) Absolute error between the real and the approximated PSNR in the interval  $[0.2, 0.8]$ . (b) Relative error

Also, it appears in Fig. 2 that, when the SSIM varies in  $[0.2, 0.8]$ , the curves are essentially comparable to straight lines (an example is given by the red line plotted for the case  $\sigma_{fg} = 10^2$ ). Computing the equation of the straight lines yields the approximated PSNR, denoted  $\text{PSNR}_{sl}$ , as follows in the interval  $[0.2, 0.8]$ :

$$\text{PSNR}_{sl} = 20.069 \times \text{SSIM} + (10 \log_{10}(255^2 / 2\sigma_{fg}) - 10.034) \quad (11)$$

In Fig. 3, we plot the absolute error ( $\Delta P = \text{PSNR} - \text{PSNR}_{sl}$ ) and the relative error ( $|\Delta P|/\text{PSNR}$ ) of the approximation. As can be observed, the maximum relative error is only 0.8 %, which indicates that the linear approximation is accurate enough.

### III. EXPERIMENTAL RESULTS

The relationship derived so far between the SSIM and the PSNR is quite interesting, but does not actually indicate if one measure is more or less sensitive to any image degradation than the other. Thus, we have no information on how the values of the PSNR and the SSIM are influenced by any degradation applied to images. For this purpose, comparisons of the PSNR and the SSIM values based on experiments using various original and degraded images are generally required [2,8,9,12]. In this paper, we use F-scores to measure the sensitivity of the PSNR and the SSIM. More concretely, we measure how the PSNR and the SSIM are influenced by the parameters of the Gaussian noise, Gaussian blur, jpeg and jpeg2000 compression respectively, which were presented in Section 2. The images used for the experiments come from the Kodak database, shown in Fig. 1. To define the F-score, let us consider a set of parameter values of a given image degradation (for example, quality parameters of jpeg compression=30%, 50%, 70%, 90%). For each parameter, different values of the PSNR, forming a group, are computed for the original images. The same is made for the SSIM. The F-score associated to the PSNR corresponds to the ratio of the variance of the set of mean values of the PSNR in all groups over the mean value of the within-group variances. The F-score of the SSIM is computed similarly. The F-score varies in  $[0, \infty[$ : a low value indicates that the parameters do not have a great impact on the values of the quality measure, meaning a low sensitivity of the quality measure to the parameters; a high value of the F-score, on the contrary, indicates a great impact of the parameters on the values of the quality measure, meaning a high sensitivity. A similar approach was used in [2] to compare different quality measures.

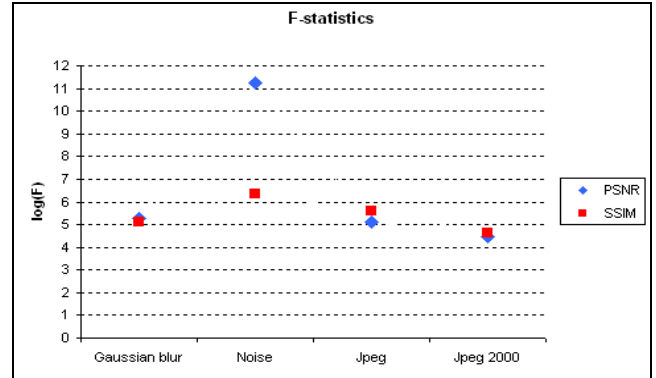


Figure 4. Comparison of the sensitivity of the PSNR and the SSIM using the F scores.

In Fig. 4, we present the results of the F-score for the various degradations. As can be observed, the SSIM seems to be more sensitive to jpeg compression compared to the PSNR, while the opposite is observed for additive Gaussian noise degradation. In fact, it is quite difficult to find a quality measure that is more sensitive to additive Gaussian noise than the PSNR, and some authors have noticed that in their experiments [2]. Still in Fig. 4, it appears that the SSIM is slightly more sensitive than the PSNR in discriminating the quality parameter of the jpeg2000 compression, while the PSNR is slightly better than the SSIM in discriminating the Gaussian blur. Finally, we note that the SSIM and the PSNR are more sensitive to noise degradation than all the other degradations tested in this paper. Thus, it appears that the various structural distortions introduced by additive noise are the most distinguishable for both the PSNR and the SSIM compared to the distortions introduced by Gaussian blur, jpeg and jpeg2000 compression.

### IV. CONCLUSION

In this paper, we have undertaken a theoretical study to compare the PSNR and the SSIM quality metrics by analysing their analytical formula. The study has revealed that a simple analytical link exists between the PSNR and the SSIM, which works for common degradations such as Gaussian blur, additive Gaussian noise, jpeg and jpeg2000 compression. We have also undertaken an experimental study in order to assess the sensitivity of the PSNR and the SSIM to these degradations, that is how the values of the parameter associated to each of these degradations affect the values of the PSNR and the SSIM. The study has revealed that the PSNR is more sensitive to additive Gaussian noise than the SSIM, while the opposite is observed for jpeg compression. Both measures have slightly similar sensitivity to Gaussian blur and jpeg2000 compression. In all cases, we have observed that the PSNR and the SSIM are more sensitive to additive Gaussian noise than Gaussian blur, jpeg and jpeg2000 compression.

As a final conclusion, it appears that the values of the PSNR can be predicted from the SSIM and vice-versa. The PSNR and the SSIM mainly differ on their degree of sensitivity to image degradations.

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